

# Emir Ceyani, Ph.D.

U.S. Permanent Resident

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## Current Employment

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### University of California, Los Angeles

Los Angeles, CA, USA

Postdoctoral Researcher in Computer Science | Hosts: Wei Wang & Yizhou Sun

Sep 2026 – Present

## Research Interests

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- Foundation Models
- Reinforcement Learning
- Learning over Graphs
- AI For Science & Engineering
- Probabilistic Generative Models
- Federated Learning

## Education

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### University of Southern California

Los Angeles, CA, USA

Ph.D. in Electrical & Computer Engineering | Advisor: Salman Avestimehr

Jan 2021 – Aug 2026

### Bilkent University

Ankara, Türkiye

M.Sc. in Electrical Engineering

Jun 2018 – Dec 2020

### Bilkent University

Ankara, Türkiye

B.Sc. in Electrical Engineering

Sep 2013 – Jun 2018

## Professional Experience

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### UCLA Computer Science Department

Los Angeles, CA, USA

Postdoctoral Researcher at Scalable Analytics Institute (hosts: Wei Wang & Yizhou Sun)

Sep 2026 – Present

- Leading research efforts for scalable, secure, and private distributed/federated learning for biomedical researchers for dkNET AI project in collaboration with UCI, UCSD Health, Cornell Weil Medicine, and NIH/NIDDK.
- Leading research efforts on GNNs and graph generative models for AI4S applications.
- Building scalable probabilistic methods for healthcare and biomedical applications.

### USC John O'Brien Nanofabrication Laboratory

Los Angeles, CA, USA

AI Research & Engineering Lead Advisor

Jan 2025 – Present

- Built a GP surrogate modeling and Bayesian experiment-planning system for GaAs/InP ICP-RIE processes, extending ultra-small DOE datasets with custom acquisition functions to propose viable etch recipes under limited data conditions.
- Developed a Random Forest angle estimation model with error calibration and unified quality scoring, then closed the loop with fabrication teams to translate model outputs into actionable process recipes.
- Engineered an end-to-end LLM agentic pipeline (GPT 5.5) with multi-step tool-calling for recipe generation, Bayesian optimization, robot control, and process troubleshooting, deployed as a Librechat chatbot with intent classification, parameter safety validation, and session memory for lab use.

### University of Southern California

Los Angeles, CA, USA

Graduate Research Assistant (Advisor: Salman Avestimehr)

Jan 2021 – Aug 2026

- Researching on diversity-seeking RL and their applications to diffusion models, scientific discovery and fine-tuning.
- Developing uncertainty-aware LLM fine-tuning methods based on conformal prediction for open-ended tasks to reduce the need of calibration data and extra post-hoc calibration steps.
- Designed the privacy-preserving and federated multi-modal image synthesis framework using generative adversarial networks.
- Proposed the problem of training of graph neural networks in federated and privacy-preserving settings and released the first benchmarking study to the literature.
- Designed GNNs and LLMs to automate circuit design process including topology selection and layout optimization

### TensorOpera AI (formerly FEDML AI)

Los Angeles, CA, USA

Research Scientist Intern (Supervisor: Aiden Chaoyang He)

May 2022 – Aug 2022

- Developed a secure & private platform for training federated GNNs using FedML library.
- Developed link-level differential privacy algorithm to enable link-level privacy for subgraph federated learning.

### Turkcell Technologies

Istanbul, TR

Research & Development Engineer (Supervisor: Salih Ergut)

Oct 2018 - Dec 2020

- Deployed ConvLSTM models for traffic estimation over base stations.

## Honors & Awards

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| <b>Top Reviewer Award at NeurIPS'25</b><br><i>Top Reviewer for NeurIPS'25. Awarded with conference registration</i>                                                            | 2025        |
| <b>Finalist for Qualcomm Innovation Fellowship, North America Branch</b><br><i>Abstract Title: AI Powered Revolution: Automating Analog and Radio Frequency Circuit Design</i> | 2025        |
| <b>SIAM Travel Award</b><br><i>Travel Award for SIAM-SDM'25 conference</i>                                                                                                     | 2025        |
| <b>PRAIRIE AI Accommodation &amp; Travel Award</b><br><i>Travel &amp; accommodation grant for PRAIRIE AI 2019 summer school.</i>                                               | 2019        |
| <b>Deep Bayes Accommodation &amp; Travel Award</b><br><i>Travel &amp; accommodation grant for Deep Bayes 2019 summer school.</i>                                               | 2019        |
| <b>5G and Beyond Joint Graduate Support Program Recipient</b><br><i>5G and Beyond Joint Graduate Support Program during M.Sc. studies (First recipient).</i>                   | 2018 - 2020 |
| <b>Bilkent University Graduate Fellowship</b><br><i>Awarded full tuition waiver and a stipend during the MS.c. studies at Bilkent.</i>                                         | 2018 - 2020 |
| <b>Bilkent University Graduation Awards</b><br><i>Awarded for Research Excellence Award.</i>                                                                                   | 2018        |

## Invited Talks & Poster Presentations

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### Federated Learning with Generative Models

- Invited talk at STAI Lab, Stanford University (Apr 2026)

### Three Modern Pillars of AI4Science with Graphs: Federation, Domain Knowledge, & Discovery

- Invited talk at Anima Anandkumar's AI + Science Group, Caltech CMS (Jan 2026)

### AI For Analog & RF Circuit Design

- Poster at NeurIPS'25 ML4PS & AI4Science Workshops (Dec 2025)
- Poster at the NeurIPS (Main Track) Conference (Dec 2025)
- Invited Talk at the Qualcomm Innovation Fellowship NA Finals (Apr 2025)

### Federated MRI Reconstruction with Deep Generative Models

- Poster at the ISMRM Conference (Jun 2023)
- Poster at the IEEE ISBI Conference (Apr 2023)
- Oral presentation at the Medical Imaging meets NeurIPS Workshop (Dec 2022)

### Personalized Federated Learning of Graph Neural Networks

- Oral presentation at SIAM-SDM conference (May 2025)
- Poster at the 14<sup>th</sup> USC MHI Research Festival (Oct 2024)
- Poster at the 1<sup>st</sup> Real AI@USC (Apr 2022)
- Poster at AAAI conference (Jan 2022)
- Poster at ICLR-DPML Workshop (Apr 2021)
- Oral presentation at MLSYS-GNNSYS Workshop (Mar 2021)

## Publications

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### Published and Accepted Journal Publications

- [1] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, "One model to unite them all: Personalized federated learning of multi-contrast MRI synthesis." In *Medical Image Analysis*, Volume 94, May 2024, 103121.
- [2] G. Elmas, S. Dar, Y. Korkmaz, **E. Ceyani**, B. Susam, M. Ozbey, **A.S Avestimehr**, T. Cukur. "Federated Learning of Generative Image Priors for MRI Reconstruction." in *IEEE Transactions on Medical Imaging*, July 2023.

## Publications in Conference Proceedings & Workshops (Peer-Reviewed)

- [1] A. Mehradfar, X. Zhao, Y. Huang, **E. Ceyani**, Y. Yang, S. Han, H. Aghasi, & S. Avestimehr (2025). FALCON: An ML Framework for Fully Automated Layout-Constrained Analog Circuit Design. NeurIPS'25,
- [2] **E. Ceyani**, H. Xie, B. Buyukates, C. Yang, and **S. Avestimehr**. "FedGrAINS: Personalized Subgraph Federated Learning with Adaptive Neighbor Sampling." In *SIAM International Conference on Data Mining (SDM25)*, 2025. (oral presentation, acceptance rate: 26.7%)
- [3] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, "A Personalized Federated Learning Approach for Multi-Contrast MRI Translation," in 31st annual meeting of International Society for Magnetic Resonance Imaging (ISMRM), Toronto, Canada, June 2023.
- [4] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, "Personalized, Federated, And Unified MRI Contrast Synthesis," in IEEE 20th International Symposium on Biomedical Imaging (ISBI), Virtual Conference, Apr. 2023. (Presented online)
- [5] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, "pFLSynth: Personalized Federated Learning of Image Synthesis in Multi-Contrast MRI," in NeurIPS Medical Imaging Meets, Virtual Conference (oral), Dec. 2022. (Presented online)
- [6] **E. Ceyani\***, C. He\*, K. Balasubramanian\*, M. Annavaram, and A.S. Avestimehr, "SpreadGNN: Serverless Multi-task Federated Learning for Graph Neural Networks," accepted to *AAAI*, 2022. (acceptance rate of 15%) (co-first)
- [7] **E. Ceyani\***, C. He\*, K. Balasubramanian\*, C. Yang, H. Xie, L. Sun, L. He, L. Yang, P. S. Yu, Y. Rong, P. Zhao, J. Huang, M. Annavaram, and A.S. Avestimehr, "FedGraphNN: A Federated Learning System and Benchmark for Graph Neural Networks," in *DPML workshop at ICLR and GNNSys workshop at MLSys*, 2021.
- [8] T. Ergen, and **Emir Ceyani**. "A highly efficient recurrent neural network architecture for data regression." 2018 26th Signal Processing and Communications Applications Conference (SIU). IEEE, 2018.

## Working Papers

- [1] **E. Ceyani**, X. Zhu, C. Yang, & S. Avestimehr "Reward-Driven Graph Synthesis for Cross-Silo Federated Learning Using GFlowNets,"
- [2] **E. Ceyani**, B. Buyukates, C. Yang, & S. Avestimehr "pFedFlow: Federated Learning via GFlowNet-Sampled Collaboration Graphs,"
- [3] **E. Ceyani**, A. Mehradfar, Y. Huang, H. Aghasi, & S. Avestimehr "CircGFN: Generating Realistic Analog Circuits with Generative Flow Networks"

## Professional Service

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### Program Committee Member

2022 – Present

- 1st International Workshops on Federated Learning with Graph Data at CIKM 2022 (FedGraph 2022)
- 2nd International Workshops on Federated Learning with Graph Data at IEEE ICDM 2023 (FedGraph 2023)
- CrossFL Workshop on Cross-Community Federated Learning: Algorithms, Systems and Co-designs at MLSys 2022

### Peer Reviewer

2022 – Present

- **Conferences:** ICLR'24, NeurIPS'22,'23,'25(Main & Dataset Tracks), AAAI'26,KDD'26 \*AI For Science Track)
- **Journals:** IEEE Transactions of Neural Networks and Learning Systems, IEEE Transactions of Big Data

### Summer Schools Attended

2019 – Present

- LOGML'22 Summer School, ProbAI 2021 Summer School, PAISS Summer School'19, Deep|Bayes'19 Summer School

## Teaching Experience

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### University of Southern California,

Los Angeles, CA, United States

Graduate Teaching Assistant

Aug 2023 – May 2025

- Head Teaching Assistant for EE364: Introduction to Probability & Statistics for Electrical Engineers  
Fall 2024 (Instructor: Salman Avestimehr), Spring 2025 (Instructor: Michael Neely)
- Head Teaching Assistant for EE547: Applied Cloud Computing for Electrical Engineers  
Fall 2023 (Instructor: Brandon Franzke)

### ProbAI Summer School

Remote

Voluntary Teaching Assistant

Summer 2021

- Helping participants' coding and theory related questions. Topics include probabilistic programming, variational inference, semi-implicit variational inference, normalizing flows, and neural ODEs.

### Bilkent University

Ankara, TR

Graduate Teaching Assistant

Sep 2018 – Dec 2020

- Grader for EEE443/543 - Neural Networks (Instructor: Prof. Dr. Tolga Cukur)
- Teaching Assistant for EEE 424- Digital Signal Processing (Instructor: Prof. Dr. Suleyman Serdar Kozat)
- Teaching Assistant for EEE 321 - Signals & Systems (Instructor: Prof. Dr. Haldun M. Ozaktas)
- Teaching Assistant for EEE 493/494 - Industrial Design Project I/II(Instructor: Dr. Mehmet Alper Kutay)

## Technical Skills

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**Programming Languages:** Python, Matlab, Julia, C/C++, SQL

**Developer Tools:** VS Code, Jupyter

**LLM APIs:** OpenAI, Anthropic, Gemini

**Technologies/Frameworks:** Linux, PyTorch(Geometric, Opacus), JAX, AWS, DSPy, Spark, Huggingface, vLLM